22 April 1960
SUPERSEDING
NAVORD OS 8099A
10 December 1957

MILITARY SPECIFICATION

CONNECTOR, PRACTICE WEAPON, MARK 22 MOD 0

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 This specification establishes the requirements for the procurement of the Connector, Practice Weapon, Mark 22 Mod 0, Bureau of Naval Weapons LD 479307. The Practice Weapon Connector Mark 22 Mod 0 is of one type and one class.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-354	Temperature and Humidity Test for Use in Production for Fuzes

DRAWINGS

BUREAU OF NAVAL WEAPONS

LD 479307	Connector, Practice Weapon, Mark 22 Mod 0
LD 299790	Container, Shipping for Connector, Practice Weapon Mark 22 Type

and all documents listed therein.

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PUBLICATIONS

BUREAU OF ORDNANCE

NAVORD OP 400 — General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment

(Copies of specifications, standards, drawings, and publications requested by the contractor in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on the date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATIONS

Handbook H28 and 1950 supplement - Screw Thread Standards for the Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Description. - The Practice Weapon Connector Mark 22 Mod 0 is a current transmitting device for simulating the mechanical and electrical functions of Weapon Connector Mark 21 Mod 0 and the electrical function of the Arming-Safing Switch MC-725. It also provides a means for introduction of electrical faults into the system for training purposes.

3.2 General requirements

3.2.1 Pre-production samples. - Unless otherwise specified in the contract or order, pre-production samples of the Practice Weapon Connector Mark 22 Mod 0 shall be manufactured using the methods and procedures proposed for the production lot. These samples will be tested as specified in section 4 herein and are for the purpose of determining that the production item meets the requirements of this specification and the design. Provision of samples shall be in accordance with 4.1.2.

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3.2.2 Materials. - Unless otherwise specified in the contract, requisition or order, all materials and purchased parts entering into the final assembly of the Practice Weapon Connector Mark 22 Mod 0 shall conform strictly to the detailed requirements indicated on the drawings and specifications listed in BUORD LD 479307.

3.2.3 Finishes, coatings, construction, and dimensions. - Finishes, coatings, construction, and dimensions for the Practice Weapon Connector Mark 22 Mod 0 shall be strictly in accordance with the applicable drawings, specifications, and publications listed in BUORD LD 479307.

3.2.4 Markings. - Marking of assemblies, sub-assemblies and parts shall be in accordance with the applicable drawings listed in BUORD LD 479307.

3.2.5 Interchangeability. - Component parts of assembled units (unless otherwise specified on the applicable drawings) fabricated in strict accordance with their applicable drawings shall be interchangeable and shall assemble with mating components of the unit without selection or fitting. If, during the assembly of the fabricated parts of a unit, selection or fitting is required and changes in dimension are necessary, no changes in the parts as delineated on applicable drawings will be made without prior revisions and correction of the applicable drawings. See 3.2.9.

3.2.6 Gages. - The contractor shall provide himself with whatever gages are necessary and adequate to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings. If so stipulated in the contract, order or requisition, the Government will furnish drawings of the pertinent Navy Final Inspection Gages for guidance in the design of the contractor's inspection gages. However, such procedure shall not serve to relieve the contractor of his responsibility, in the design and manufacture of such gages as may be required in the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Navy Final Inspection Gages.

3.2.7 Threads. - Unless otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28 and Supplements thereto. The class of fit for threads shall be as specified on the applicable drawings, and the NOT-GO gaging practice shall be as set forth in the supplement to the aforementioned Handbook H28.

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3.2.8 Requests for deviation. - Requests for deviation from this specification, applicable drawings, specifications, publications, materials and processes specified herein, shall be as set forth in OP 400.

3.2.9 Conflicting requirements. - Conflicting requirements arising between this specification or any specifications, publications, or drawings listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation and clarification.

3.3 Definitions. - For the purpose of this specification, the following definitions are applicable:

3.3.1 Practice weapon connector. - Reference to practice weapon connector, unless otherwise indicated, shall mean Practice Weapon Connector Mark 22 Mod 0 as covered by BUORD LD 479307.

3.4 Performance requirements and product characteristics. - The practice weapon connector shall meet the following performance requirements and product characteristics:

3.4.1 Circuit resistance. - When tested in accordance with 4.5.4.1, the electrical resistance of each practice weapon connector circuit shall be less than 0.10 ohm, unless otherwise specified in test paragraph.

3.4.2 Insulation resistance. - When tested in accordance with 4.5.4.2, the practice weapon connector insulation resistance shall be at least 20 megohms for each indicated circuit.

3.4.3 Static adapter plug disconnect force. - The ball retainer assembly shall start movement and the adapter plug shall release from the practice weapon connector when tested as specified in 4.5.4.3. There shall be no evidence of damage, jamming or misalignment of parts after release.

3.4.4 Ball retainer assembly fit. - The practice weapon connector with the adapter plug removed shall engage smoothly with acceptance gage. There shall be no jamming, roughness in turning or alignment difficulty encountered during coupling or uncoupling of this equipment with the ball retainer assembly.

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3.4.5 Adapter plug fit. - The adapter plug shall engage with acceptance gage and rotate to the locked position. There shall be no binding or roughness when rotating the adapter locking sleeve (1476515) from the locked to the unlocked position or reverse.

3.4.6 Motor driven switch functioning. - The motor driven switch running time shall be tested in accordance with 4.5.4.4. The time required to complete one half cycle (SAFE to ARMED positions) shall be 4 to 7 seconds. The motor steady running current shall not exceed 1.0 ampere.

3.5 Environmental and durability requirements.

3.5.1 Aircraft vibration. - The practice weapon connector shall be subjected to the aircraft vibration test conditions specified in 4.6.2 and thereafter shall show no damage and meet the performance requirements and the product characteristics of this specification.

3.5.2 Transportation vibration. - The practice weapon connector shall be subjected to the transportation vibration test conditions specified in 4.6.3 and thereafter shall show no damage and meet the performance requirements and the product characteristics of this specification.

3.5.3 Extended operation. - The practice weapon connector shall be subjected to the extended operation test conditions specified in 4.6.4 and thereafter shall meet the performance requirements and product characteristic of this specification.

3.5.4 Low temperature operation. - The practice weapon connector shall be subjected to low temperature operating condition specified in 4.6.5 and thereafter shall show no damage and meet the performance requirements and product characteristic of this specification.

3.5.5 Temperature and humidity. - The practice weapon connector shall be subjected to temperature and humidity conditions specified in 4.6.6 and thereafter shall meet the performance requirements and the product characteristics of this specification.

3.6 Clean-up. - Prior to and after final assembly all parts, components and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. Burrs and sharp edges as well as rosin flash shall be removed.

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3.7 Workmanship. - The equipment, including all parts and accessories, shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, machine screw assemblage, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Government inspection. - The Government inspector will make such inspections as are necessary to determine that components and assemblies are in accordance with the requirements of the contract, pertinent drawings and specifications. Unless otherwise specified, these inspections will be conducted in accordance with instructions contained in OP 400.

4.1.1 Sampling. - Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government inspector in the determination of the acceptability of products submitted by a supplier for Government inspection shall be in accordance with the provisions set forth in MIL-STD-105.

4.1.2 Pre-production sample. - Unless otherwise specified for each contract or order, a representative sample of four (4) practice weapon connectors shall be manufactured using the methods and procedures proposed for the production lot. These shall be submitted to the Government inspector for pre-production tests at an activity designated in the contract or by the Bureau of Ordnance, for purposes of determining compliance with the requirements of the contract, specifications, and drawings. Further production of the item by the contractor prior to approval of the contracting activity or the completion of the tests on pre-production samples shall be at the contractor's risk. Pre-production samples accepted will be supplied as part of the quantity specified by the contract or order.

4.1.3 Lot size. - As applied to Government inspection of units of product the term "lot" shall mean "inspection lot", i.e. a collection of units of product submitted by a supplier for Government inspection. Unless otherwise specified, the number of units of product in "inspection lots" will be determined by the Government inspector and may differ from the quantity designated in the contract or order in a lot for production, shipment, or other purposes.

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4.2 Contractor's inspection. - Contractor's inspection shall be conducted in accordance with the provisions of "Inspection by contractor" as contained in NAVORD OP 400.

4.3 Classification of tests. - The inspection and testing of the practice weapon connector shall be as follows:

4.3.1 Acceptance tests. - The acceptance tests are to be accomplished on the practice weapon connector being submitted for acceptance under contract. Inspection and test procedures shall be submitted for Government approval prior to commencement of production. Acceptance tests shall either be performed by the manufacturer and witnessed by the Government inspector, or shall be conducted by the Government inspector. These tests are defined in 4.5.

4.3.2 Pre-production tests. - Pre-production tests are those which are accomplished on samples selected as specified in 4.1.2 which are representative of the production of the practice weapon connector after award of the contract, to determine that the lot and production meet the requirements of this specification. These are detailed in 4.6. Acceptance shall be based on no defects in a sample. Failure of a sample to comply with these requirements will result in the rejection of the lot or the cessation of production as determined by the procuring activity.

4.4 Test equipment. - The following items of test equipment are required to perform the acceptance tests set forth in this specification:

4.4.1 Megohmmeter 500 volts d-c, 10-200 megohms ± 10%.

4.4.2 Resistance bridge, 0.01 - 1.0 ohm ± 2%.

4.4.3 Timer clock accurate to within ± 1%.

4.4.4 Ammeter, Center "O", 0-5 amperes ± 10%.

4.4.5 Spring load, dead load tester, 0-100 ± 10%.

4.4.6 Dial indicator, 0-1 ± 0.001 inch.

4.4.7 Test Set, Weapon Connector Mk 21 Mod 0, adapter plug static disconnect force, BUORD LD 486038 (Government furnished equipment)

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4.5 Acceptance tests.

4.5.1 The Contractor or the Government inspector shall conduct acceptance tests as specified in 4.3.1 to assure that the practice weapon connector is in compliance with the requirements of 3.2 and 3.4.

4.5.2 Test conditions. - Unless otherwise specified the test procedure of 4.5.4 shall be accomplished under the following conditions:

4.5.2.1 Temperature:	Room ambient 65 degrees F to 95 degrees F (18.33 degrees C to 35 degrees C).
4.5.2.2 Altitude:	Normal ground.
4.5.2.3 Vibration:	None
4.5.2.4 Humidity:	Room ambient to 95 percent relative maximum.

4.5.3 Test and inspection equipment and facilities. - The manufacturer shall furnish and maintain all necessary test equipment, facilities, and personnel for performing all acceptance tests. The test equipment shall be adequate in quantity, and when definite requirements are not specified they shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

4.5.4 Test procedure.

4.5.4.1 Circuit resistance test. - With the adapter plug installed, and the selector switch in position #1, place the motor driven switch in the SAFE position by connecting the positive terminal of a 28 volt d-c source to contact T and the negative terminal to contact B. Run the motor until it stops. Using resistance bridge (4.4.2), measure the electrical resistance between the contact points listed in each of the following phases of testing. Practice weapon connectors having resistance not greater than 0.10 ohm through each of the indicated circuits, unless otherwise specified, shall be considered to have met the specified requirements.

4.5.4.1.1 With the selector switch in position #1, measure the circuit resistance between the following contact points:

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Adapter plug 1519862	Cable assembly plug 1541079
H-J-G	
V	B
L-U	
S-K-N	
E	C
A-F	
P	A

4.5.4.1.1.1 With the selector switch in each of positions #1, #3, #4 and #5, measure the circuit resistance between the following contact points. This resistance shall be 15 ± 10 ohms:

Adapter plug
C-R
B-R

4.5.4.1.2 With the selector switch in each of positions #2, and #3, measure the circuit resistance between the following contact points:

Adapter plug	Cable assembly plug
V	B
E	C

4.5.4.1.3 With the selector switch in position #4, measure the circuit resistance between the following contact points:

Adapter plug	Cable assembly plug
V	B

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4.5.4.1.4 With the selector switch in position #5, measure the circuit resistance between the following contact points:

Adapter plug	Cable assembly plug
E	C

4.5.4.1.5 With the selector switch in position #1, place the motor driven switch in the ARMED position by connecting the positive terminal of a 28 volt d-c source to contact B and the negative terminal to contact R. Run the motor until it stops. Using low resistance bridge listed in 4.4.2, measure the electrical resistance between the following contact points:

Adapter plug

A-D

4.5.4.1.5.1 With the selector switch in each of positions #1, #2, #4 and #5, measure circuit resistance between the following contact points. This resistance shall be 15 ± 10 ohms:

Adapter plug

B-T

C-T

4.5.4.2 Insulation resistance test. - With the adapter plug (1519862) installed, condition the practice weapon connector for a period of 24 hours at a relative humidity of 75 ± 5% and a temperature of 75° ± 5°F. Immediately after conditioning and with the selector switch in position #1, place the motor driven switch in the SAFE position by connecting the positive terminal of a 28 volt d-c source to contact T and the negative terminal to contact B, and run the motor until it stops. Using megohmmeter (4.4.1), measure the insulation resistance under 500 volts d-c potential in each of the following phases of testing. Practice weapon connectors having resistances of at least 20 megohms through each indicated circuit shall be considered to have met the specified requirements.

4.5.4.2.1 With selector switch in each of positions #1, #3, #4, and #5, measure the insulation resistance between each individual contact and all other contacts tied together and grounded. The following groups of contacts: H, J, and G; L and U; S, K, and N; A and F; and B, C, and R, are inter-connected and shall be considered as single contacts. These contacts are in the adapter plug.

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4. 5. 4. 2. 2 With selector switch in position #2, measure the insulation resistance between contact R and all other contacts tied together and grounded.

4. 5. 4. 2. 3 With selector switch in position #1, place the motor driven switch in the ARMED position by connecting the positive terminal of a 28 volt d-c source to contact B and the negative terminal to contact R in the adapter plug assembly and run the motor until it stops. Measure the insulation resistance between each individual contact and all other contacts tied together and grounded. The following groups of contacts: T, B and C; A and D; S, K and N; H, J and G; and L and U; are inter-connected and shall be considered as single contacts.

4. 5. 4. 2. 3. 1 With selector switch in position #3, measure the insulation resistance between contact T and all other contacts tied together and grounded.

4. 5. 4. 2. 4 With selector switch in position #4, measure the insulation resistance between contact C in the cable assembly plug (1541079) and all contacts of the adapter plug tied together and grounded.

4. 5. 4. 2. 5 With selector switch in position #5, measure the insulation resistance between contact B in the cable assembly plug (1541079) and all contacts of the adapter plug tied together and grounded.

4. 5. 4. 3 Static adapter plug disconnect force test. - The ball retainer assembly shall begin to move from its fully seated position when the adapter plug is subjected to a vertical pull force of 30 pounds minimum when using 4. 4. 5 and 4. 4. 6 or test set (4. 4. 7). The adapter plug shall release from the practice weapon connector, when this force is gradually increased, at 75 + 15 - 5 pounds. If the ball retainer assembly begins movement at the minimum specified pull force and the adapter plug releases within the specified limits without damage, jamming or misalignment of parts, the practice weapon connector shall be considered to have met the requirements of this test. At the conclusion of the test the adapter plug shall be reinstalled in the practice weapon connector.

4. 5. 4. 4 Motor driven switch functioning test. - The motor driven switch terminal contacts shall be connected to the motor timing test circuit as shown in Fig. 1. The motor driven switch shall be operated and the running time required to complete one half cycle (SAFE to

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ARMED) shall be measured with the motor current not exceeding 1.0 ampere. Motor driven switches operating within a running time of 4 to 7 seconds shall be considered to have met the requirements of this test.

4.6 Pre-production tests:

4.6.1 After meeting the performance requirements of 3.4, subject the four (4) practice weapon connectors selected for pre-production testing purposes, to the following tests in the order specified.

4.6.1.1 Subject the practice weapon connectors to transportation vibration test 4.6.3 and then acceptance tests 4.5.4.1, 4.5.4.2 and 4.5.4.4.

4.6.1.2 Subject the practice weapon connectors to the temperature and humidity test 4.6.6, and then acceptance tests 4.5.4.1, 4.5.4.2, 4.5.4.3 and 4.5.4.4.

4.6.1.3 Subject the practice weapon connectors to the aircraft vibration test 4.6.2 and then acceptance tests 4.5.4.1, 4.5.4.2 and 4.5.4.3, and the low temperature operation test 4.6.5.

4.6.1.4 Subject the practice weapon connectors to the extended operation test 4.6.4 and then acceptance tests 4.5.4.1 and 4.5.4.2.

4.6.2 Aircraft vibration test. - Mount the practice weapon connector with adapter plug installed in test fixture figure 3 and subject to vibration testing as follows. Securely mount the practice weapon connector in the test fixture on the vibration table in such a manner that the longitudinal axis of the weapon connector is horizontal and the axis of the adapter plug is vertical. The table shall vibrate in a plane perpendicular to the longitudinal axis of the weapon connector and parallel to the axis of the adapter plug. The acceleration of vibration will be maintained constant at 3g peak value, while the frequency is varied in one continuous sweep from 50 to 500 to 50 cps. The duration of the test is 180 minutes.

4.6.3 Transportation vibration test. - Mount the practice weapon connector with adapter plug installed in test fixture figure 3 and subject the mounted practice weapon connector to vibration testing as follows: Rigidly mount the practice weapon connector in test fixture on the vibration table in such a manner that the longitudinal axis of the weapon connector is horizontal and the axis of the adapter plug is vertical. The table shall vibrate in a plane perpendicular to the longitudinal axis of the

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weapon connector and parallel to the axis of the adapter plug. Maintain the vibration at the following single amplitude while the frequency is varied in 100 cpm steps from 700 to 3000 cpm. The duration of each frequency step is 30 minutes.

AMPLITUDE	FREQUENCY
0.030 inch (±.002)	700 to 1500 cpm
0.020 inch (±.002)	1600 to 2000 cpm
0.013 inch (±.001)	2100 to 2500 cpm
0.009 inch (±.001)	2600 to 3000 cpm

The duration of the complete test is 12 hours.

4.6.4 Extended operation test. - The practice weapon connector shall be subjected to extended operation testing as follows: Connect contact points in weapon connector to test circuit terminals as shown in test circuit Figure 2 and operate motor driven switch from SAFE to ARMED to SAFE for 1000 cycles. Motor current shall not exceed 1.0 ampere.

4.6.5 Low temperature operation test. - Immediately after conditioning at -20° F for a period of 24 hours, subject the practice weapon connector to motor driven switch functioning test 4.5.4.4.

4.6.6 Temperature and humidity test. - The practice weapon connector shall be subjected to two 14 day "JAN Temperature and Humidity Cycles" as specified in MIL-STD-354.

4.7 Acceptance criteria. - Any practice weapon connector which fails to meet any of the performance requirements, product characteristics or environmental requirements of this specification shall be regarded as defective. The inspector shall ascertain that the packaging, packing and marking of accepted items are in accordance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.

5.1.1 Level A.

5.1.1.1 Each Connector, Practice Weapon Mark 22 shall be packaged and packed in accordance with BUORD LD 299790.

5.1.2 Level C. - (Not applicable)

5.2 Marking.

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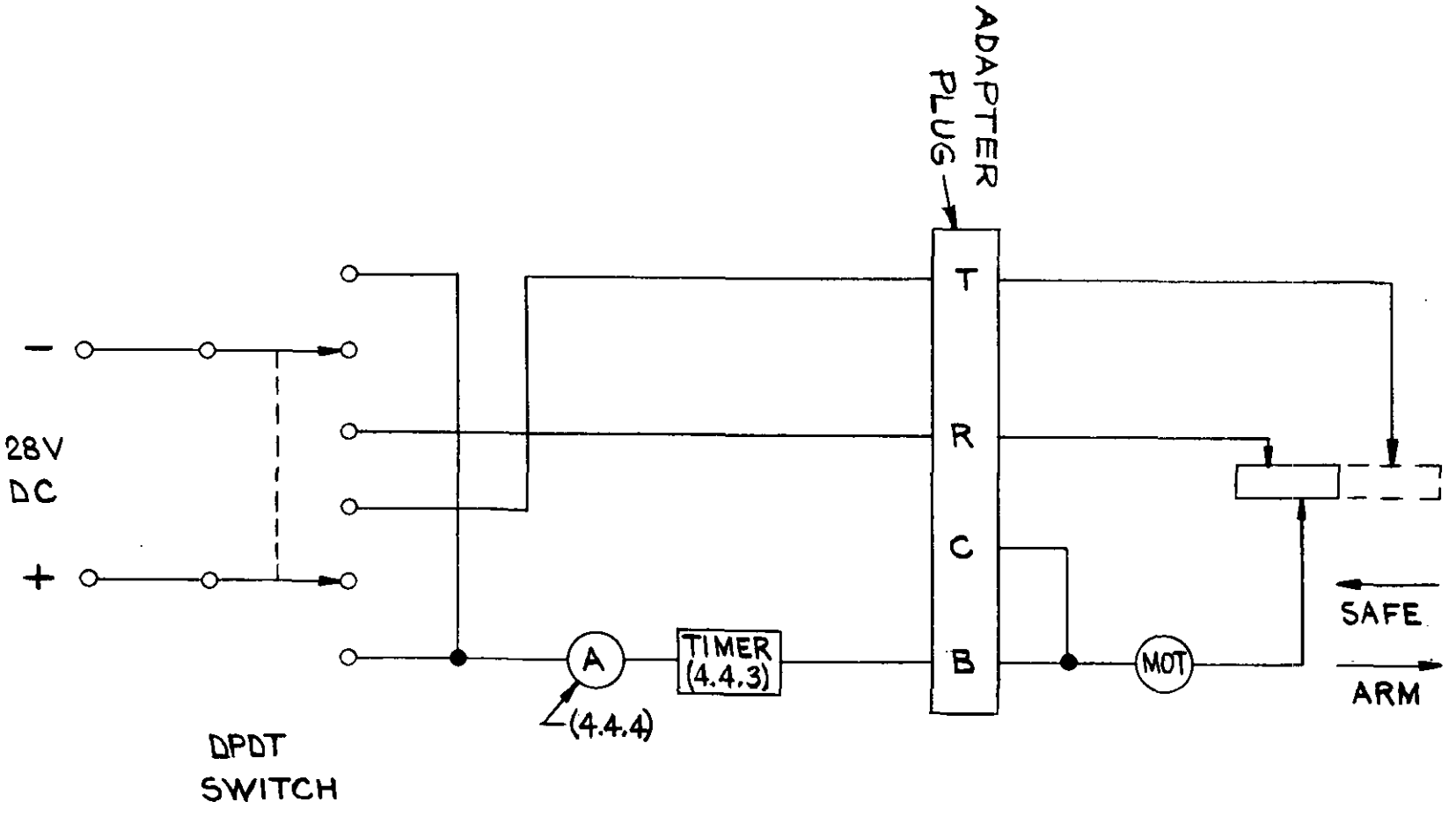
5.2.1 In addition to any special marking required by the contract or order, all marking shall be in accordance with Military Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the specific title and number of this specification and the exceptions to this specification, and the applicable drawings and other documents.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever, and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any invention that may be in any way related thereto.

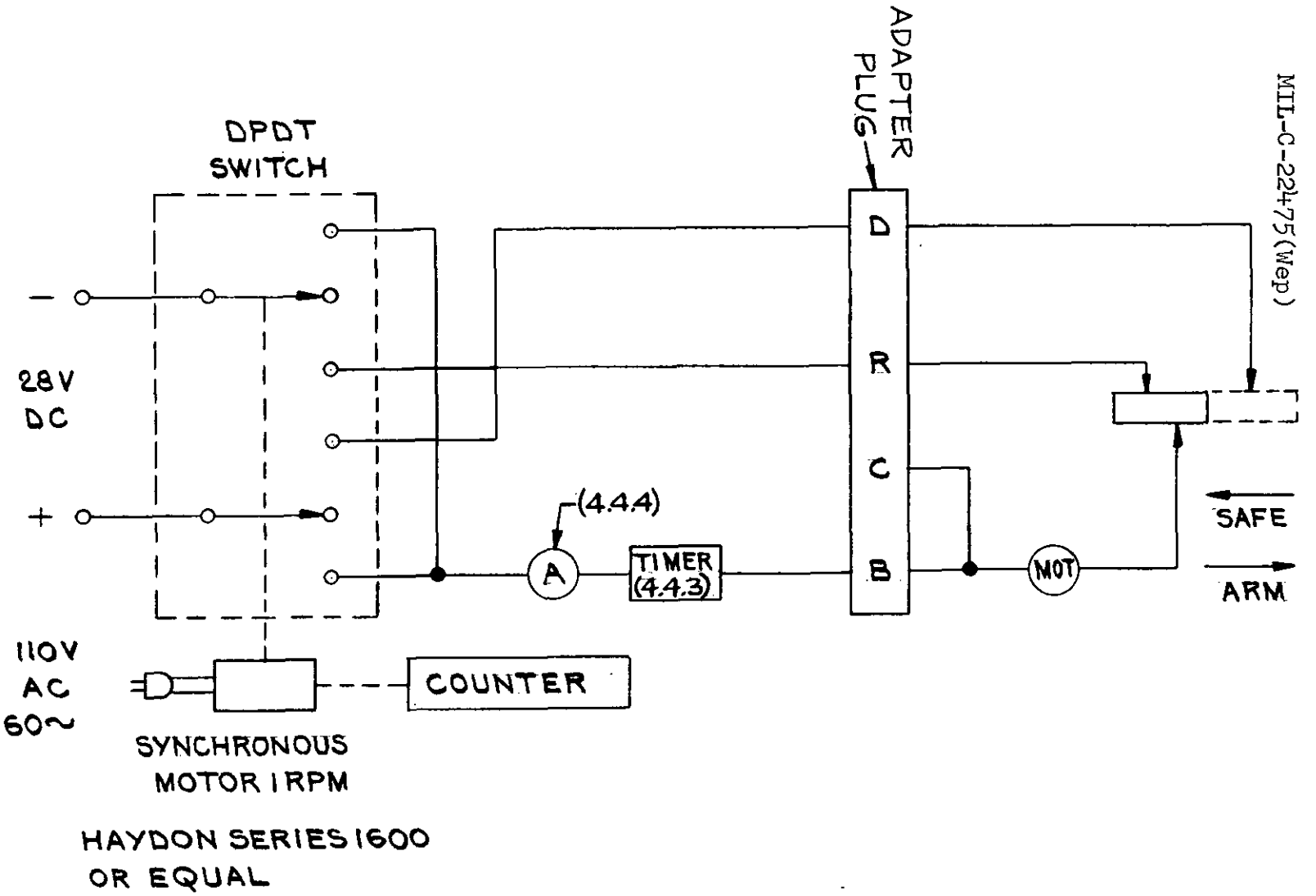
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TEST CIRCUIT

FIGURE 1

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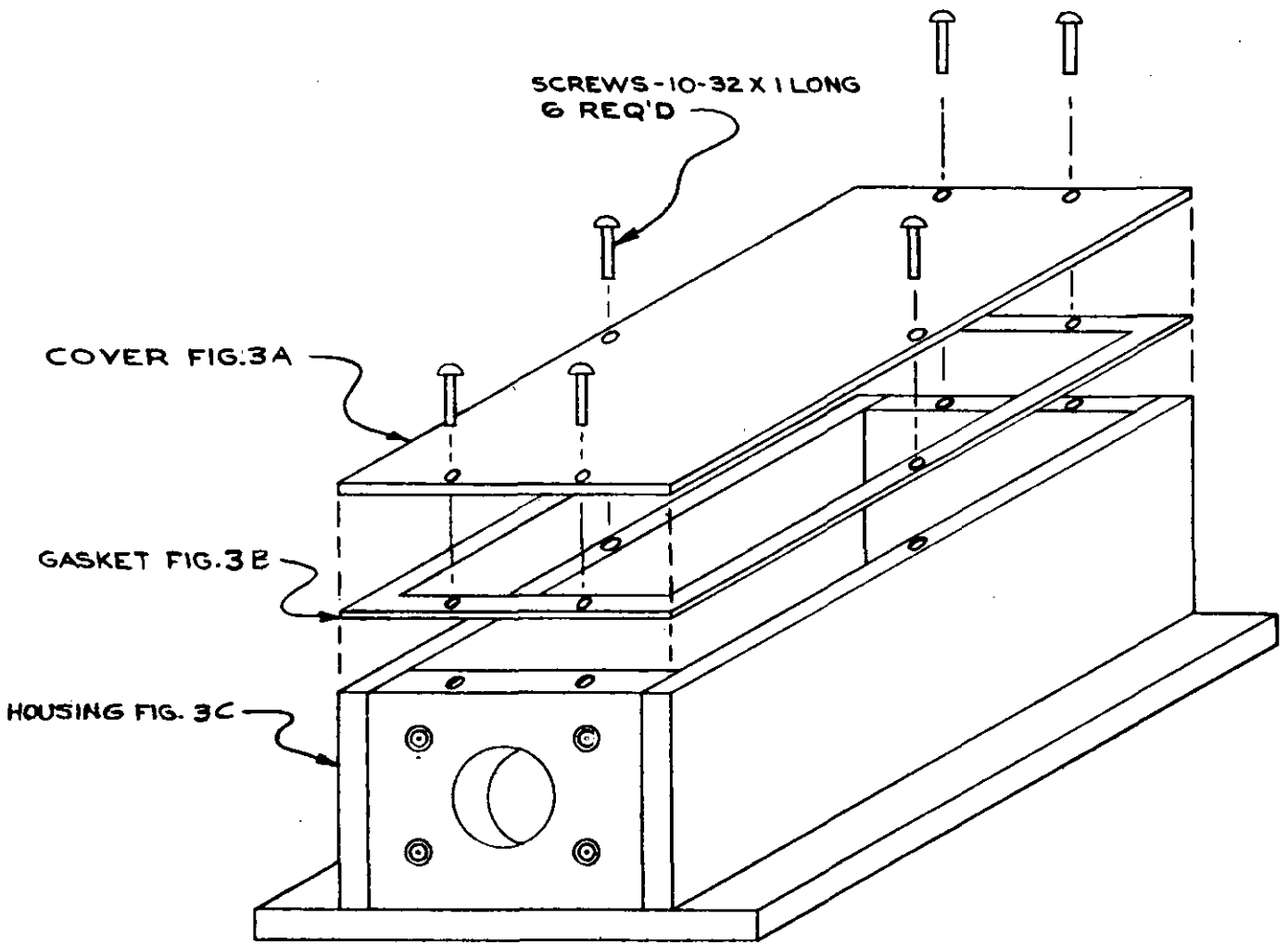


TEST CIRCUIT

FIGURE 2

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NOTE:
WEAPON CONNECTOR SHALL BE MOUNTED ON OUTSIDE OF TEST FIXTURE



TEST FIXTURE
(EXPLODED)
Figure 3

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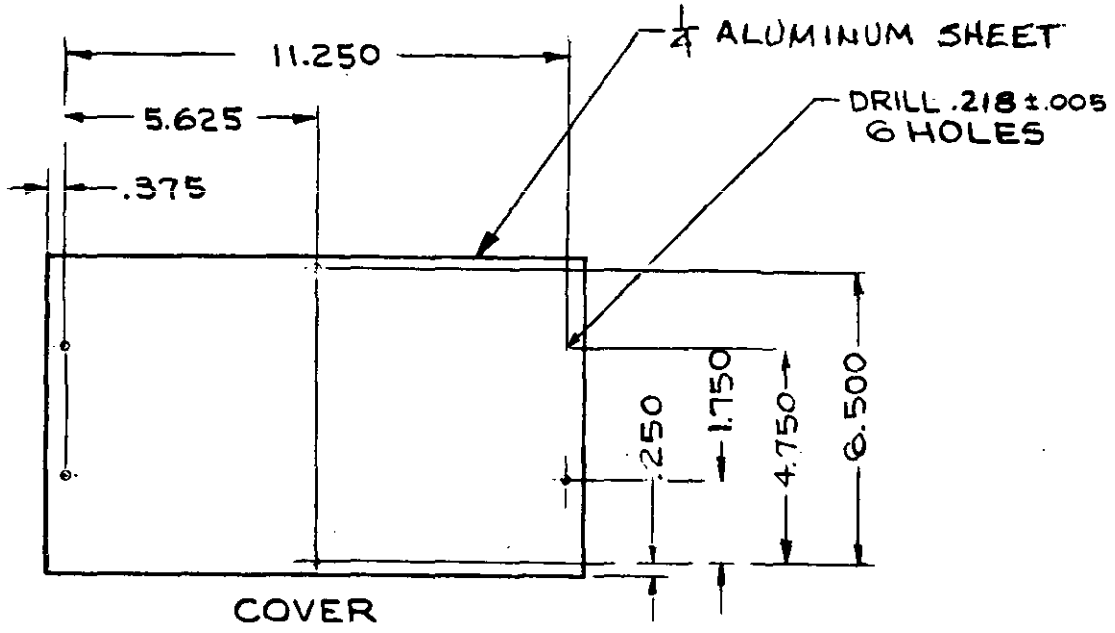


Figure 3A

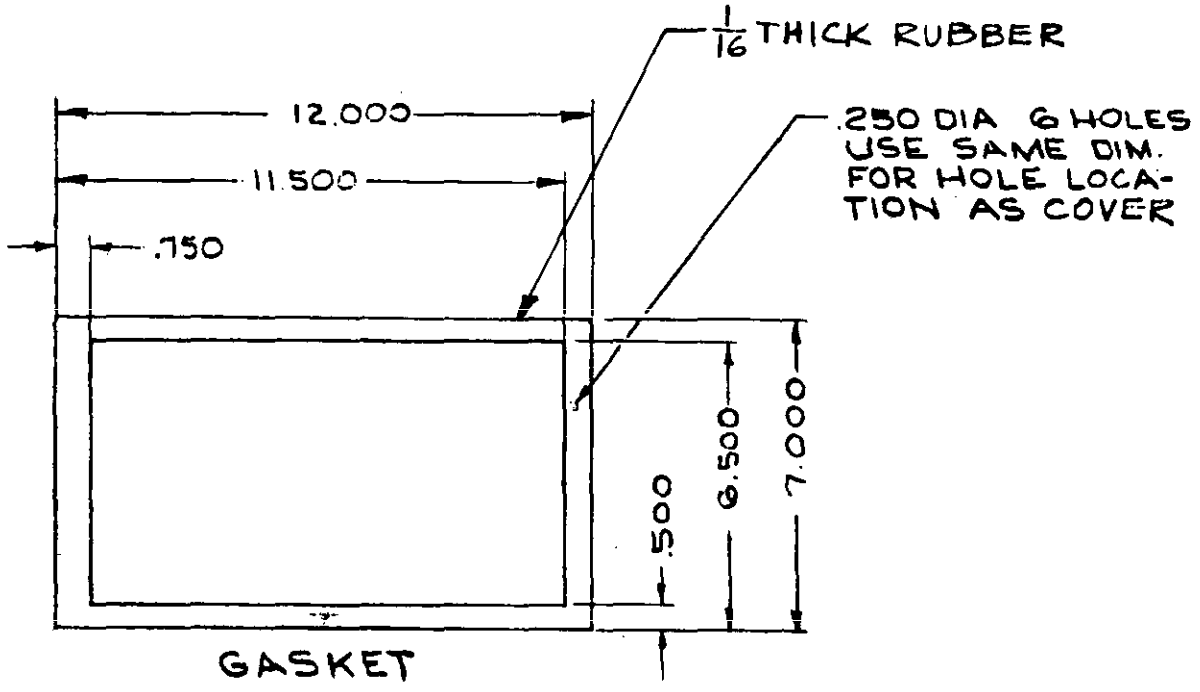
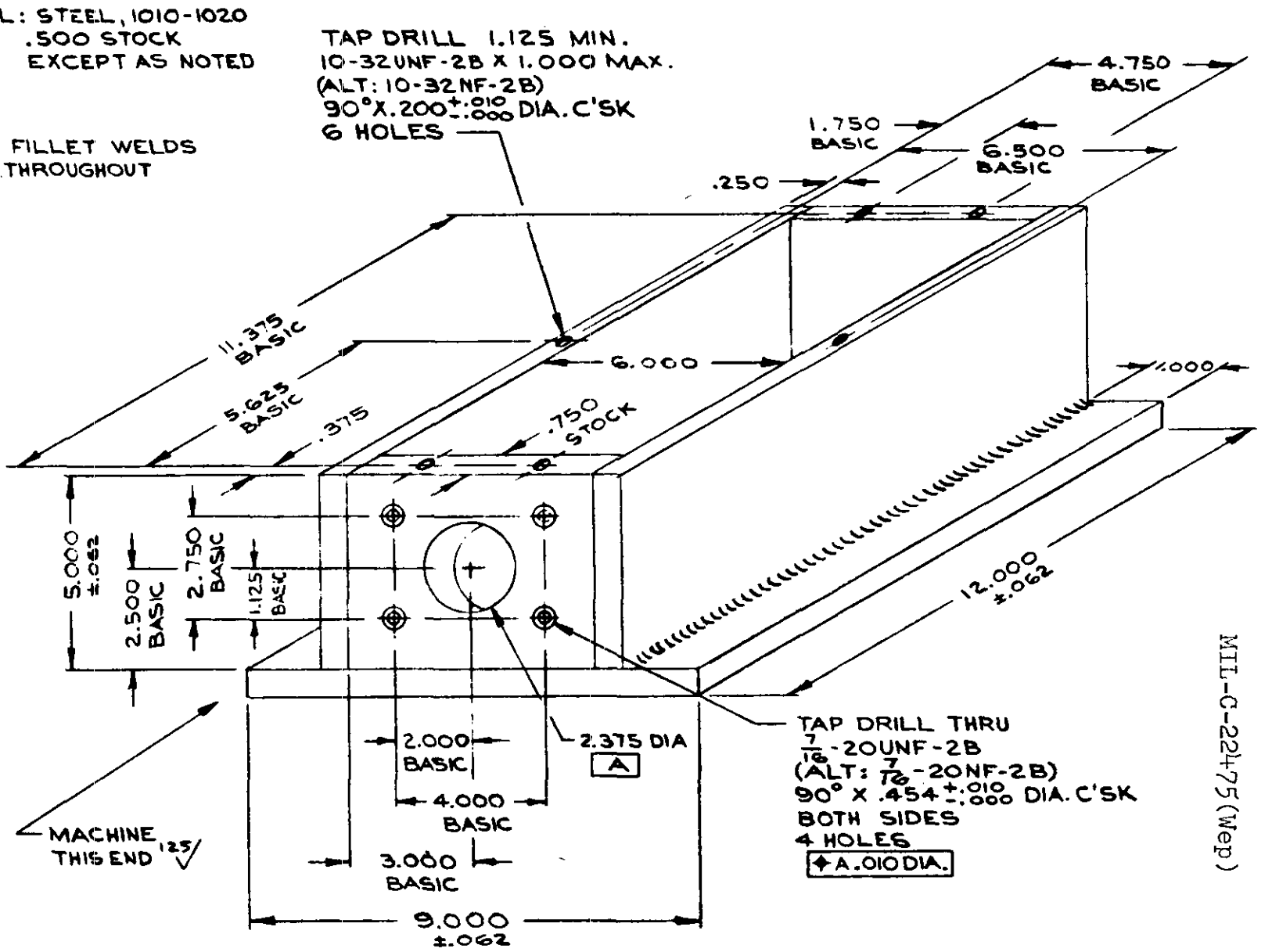


Figure 3B

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HOUSING

Figure 3C

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